

Method Pilot: Test the Test

BUILD • Science • 40 minutes

I can carry out or direct a small method pilot, identify a procedural problem and revise the plan ready for W14.

Prepare before pupils arrive

Setup: Keep final A/B samples and result sheets out of W13B. Prepare one labelled practice route plus fixed P1/P2 pilot records and method V1/V2 cards.

Safety: Local risk assessment is the authority. If the practice rig is unsealed, unstable, damaged, dusty or uninformative, do not improvise; use the fixed pilot log and approve the fixed W14 route.

Equipment: One clearly labelled staff-prepared practice rig or fixed pilot log; stable tray; 5 mL needle-free tool; 120-second timer; collector; wipes; method V1/V2 sheet; readiness card.

Safety: Only staff build and approve a practice rig. No final A/B samples are used. No hammering, chipping, scratching, licking, tasting, smelling or dust blowing. Stop for leaks, instability, damage, flaking, dust or unclear measurement. The fixed pilot-log/director route is full participation.

Fallback: Use printed pilot cards and method V1/V2 strips. An adult moves or scribes exactly as directed; no water or rock handling is required.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Pilot and revise a simple fair-test method ready for the full rock investigation.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Last lesson: what is the difference between pilot and final investigation? Last lesson: name one control the pilot should check.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	pilot — small practice run fault — a problem in the method revise — change the plan clearly Keep final Samples A and B out of the pilot. Use one clearly labelled practice sample or the fixed pilot log so today cannot become the full comparison.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Compare method version 1 with version 2. Highlight the exact changed instruction, such as check the seal all the way around before adding water.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each pilot note to KEEP, REPAIR or W14 READINESS. Do not route any card to a final rock conclusion.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Carry out or analyse one practice pilot, record a method fault or successful check, revise the method and issue a W14 readiness decision without a final rock conclusion. Record: Pilot checklist, one procedure observation or fault, method V1 and visibly revised V2, safety/access decision, W14 readiness card and a statement of what the pilot cannot conclude.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Describe the method fault, revision and readiness decision without giving a rock conclusion. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: A leak occurs, so add 'press and inspect the seal before adding 5 mL'; rerun the practice route and keep final A/B results blank.

Correct. The response repairs one method fault, retests the procedure and preserves the final investigation for W14.

Misconception repair

Repaired. Method evidence and final sample evidence have different jobs and remain in separate records.

Guided checkpoint

Correct. The response repairs one method fault, retests the procedure and preserves the final investigation for W14. Choose the option that changes the instruction, not the scientific result or several variables.

Practice label

KEEP IN METHOD V2. Keep: clear labelling prevents pilot data entering the final comparison. This feature protects the boundary between practice and final evidence.

Volume step

KEEP IN METHOD V2. Keep: another person can reproduce the starting volume. This instruction is already specific and measurable.

Stop rule

KEEP IN METHOD V2. Keep: the method states when physical work must stop. A safe procedure needs an explicit stop condition.

Pilot fault P1

REPAIR THE METHOD. Repair: add a full seal inspection before water is added; mark P1 invalid and do not infer permeability. The water bypassed the practice sample, so the method needs a seal revision.

Table fault

REPAIR THE METHOD. Repair: change the heading to Water collected (mL). Another person must know exactly what the number measures.

Timing fault

REPAIR THE METHOD. Repair: state start timer when water is added and read the collector at 120 seconds. A reproducible method needs one defined time.

Ready decision

W14 READINESS DECISION. W14 READY: use method version 2 with final matched samples and planned repeats. This card reports procedure readiness, not which final sample will pass more water.

Repair then review

W14 READINESS DECISION. REPAIR THEN REVIEW: do not start the physical final comparison; use the fixed W14 route unless the method is safely reapproved. Readiness can honestly be no; the final investigation waits for a valid route.

Fixed-route decision

W14 READINESS DECISION. W14 READY via fixed data: access and safety change the route, not the scientific reasoning goal. This card makes an approved future-route decision.

Model limitation

A successful pilot only shows that the procedure can be followed and recorded under practice conditions. It does not prove the final samples will produce a pattern or answer the investigation question.

Independent evidence

Pilot checklist, one procedure observation or fault, method V1 and visibly revised V2, safety/access decision, W14 readiness card and a statement of what the pilot cannot conclude.

Supported: Sequence the practice steps, choose one fault card and one matching repair, then select ready, review or fixed route for W14. Use: The pilot showed ___. We changed ___. For W14 we will ___.

Standard: Complete the pilot log, justify one precise revision and issue an evidence-based readiness decision. Keep method observation, revision and readiness in separate boxes; leave final conclusion blank.

Stretch: Evaluate whether the revision fixes the identified fault without creating a new variable, then specify the W14 approval conditions. Explain why a successful pilot increases procedural confidence but not confidence in a final sample pattern.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W13B_Method_Pilot_Test_The_Test_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.